Robotic Navigation and Exploration

HW4: Object Detection & Semantic Segmentation

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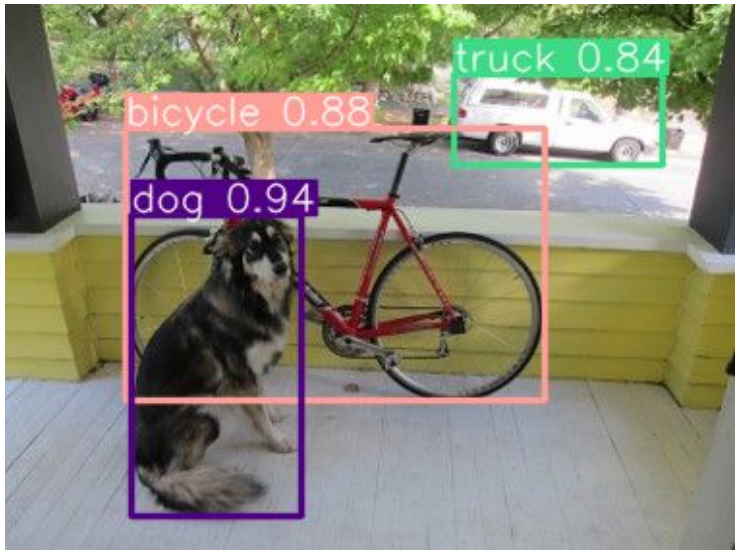
Agenda

- Introduction to YOLO
- Object Detection
- Semantic Segmentation
- Data Collection
- Data Labeling
- YOLO Training
- YOLO Testing
- Grading & Submission

Introduction to YOLO

YOLO

- YOLO: You Only Look Once
- A popular, real-time object detection and image segmentation model
- Official website: https://docs.ultralytics.com/
- Born in 2015
- Versions: YOLOv1~YOLOv12, YOLO26



YOLO26

Performance

Detection (COCO) | Segmentation (COCO) | Classification (ImageNet) | Pose (COCO) | OBB (DO

See Detection Docs for usage examples with these models trained on COCO, which include 80 pretrained classes.

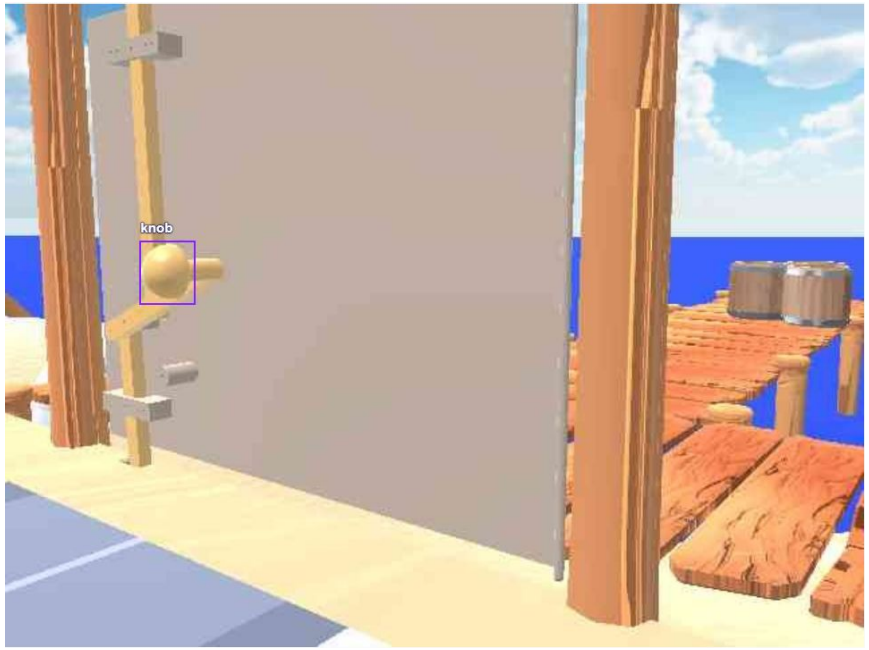
Model	size (pixels)	mAP^{val} 50-95	mAP^{val} 50-95(e2e)	Speed CPU ONNX (ms)	Speed T4 TensorRT10 (ms)	params (M)	FLOPs (B)
YOLO26n	640	40.9	40.1	38.9 ± 0.7	1.7 ± 0.0	2.4	5.4
YOLO26s	640	48.6	47.8	87.2 ± 0.9	2.5 ± 0.0	9.5	20.7
YOLO26m	640	53.1	52.5	220.0 ± 1.4	4.7 ± 0.1	20.4	68.2
YOLO26l	640	55.0	54.4	286.2 ± 2.0	6.2 ± 0.2	24.8	86.4
YOLO26x	640	57.5	56.9	525.8 ± 4.0	11.8 ± 0.2	55.7	193.9

https://docs.ultralytics.com/models/yolo26/

Object Detection

Goal

- Predict the bounding box of each object
- In this homework, you're asked to detect bear and knob



Dataset Format

zidane.txt

```
0  0.481719 0.634028 0.690625 0.713278
0  0.741094 0.524306 0.314750 0.933389
27 0.364844 0.795833 0.078125 0.400000
```

Class X Y Width Height

Semantic Segmentation

Goal

- Predict the area of each object
- In this homework, you're asked to detect road and bridge

Dataset Format

The format for a single row in the segmentation dataset file is as follows:

```
<class-index> <x1> <y1> <x2> <y2> ... <xn> <yn>
```

In this format, `<class-index>` is the index of the class for the object, and `<x1> <y1> <x2> <y2> ... <xn> <yn>` are the bounding coordinates of the object's segmentation mask. The coordinates are separated by spaces.

Here is an example of the YOLO dataset format for a single image with two objects made up of a 3-point segment and a 5-point segment.

```
0 0.681 0.485 0.670 0.487 0.676 0.487
1 0.504 0.000 0.501 0.004 0.498 0.004 0.493 0.010 0.492 0.0104
```

Data Collection

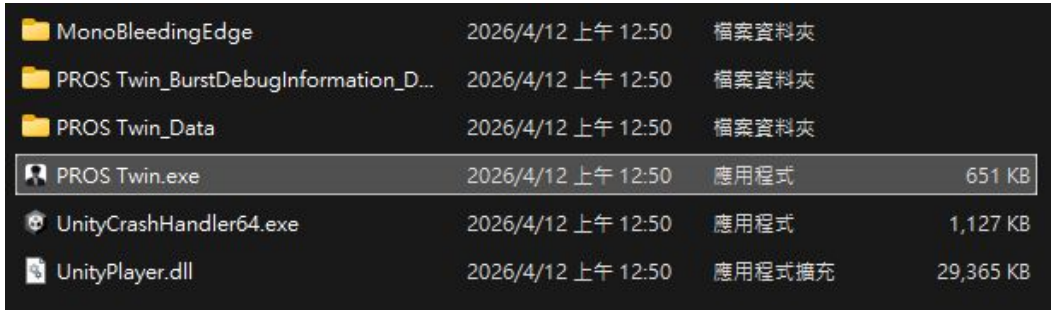
Data Collection

- Open Docker Desktop (**Never close it!!!**)
- Connect VSCode to WSL
- Execute pros_app

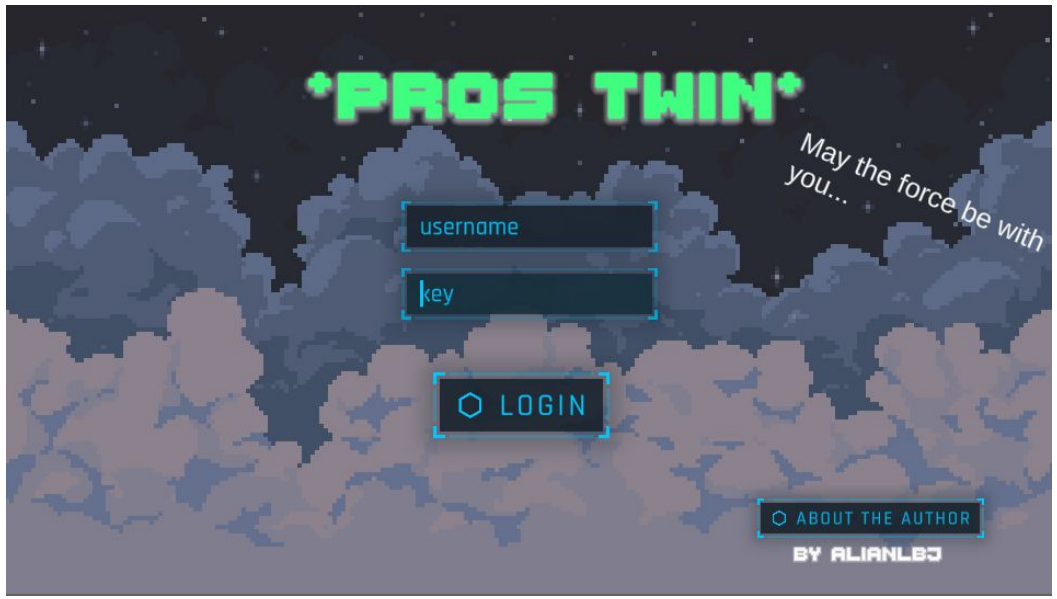
```
#Open a terminal
cd ./workspace/pros/pros_app/
python3 ./control.py -s
#Select "./slam_unity.sh"
(Do not close it)
```

Data Collection

- Open “PROS Twin.exe” and login with your account.



MonoBleedingEdge	2026/4/12 上午 12:50	檔案資料夾	
PROS Twin_BurstDebugInformation_D...	2026/4/12 上午 12:50	檔案資料夾	
PROS Twin_Data	2026/4/12 上午 12:50	檔案資料夾	
PROS Twin.exe	2026/4/12 上午 12:50	應用程式	651 KB
UnityCrashHandler64.exe	2026/4/12 上午 12:50	應用程式	1,127 KB
UnityPlayer.dll	2026/4/12 上午 12:50	應用程式擴充	29,365 KB



Data Collection

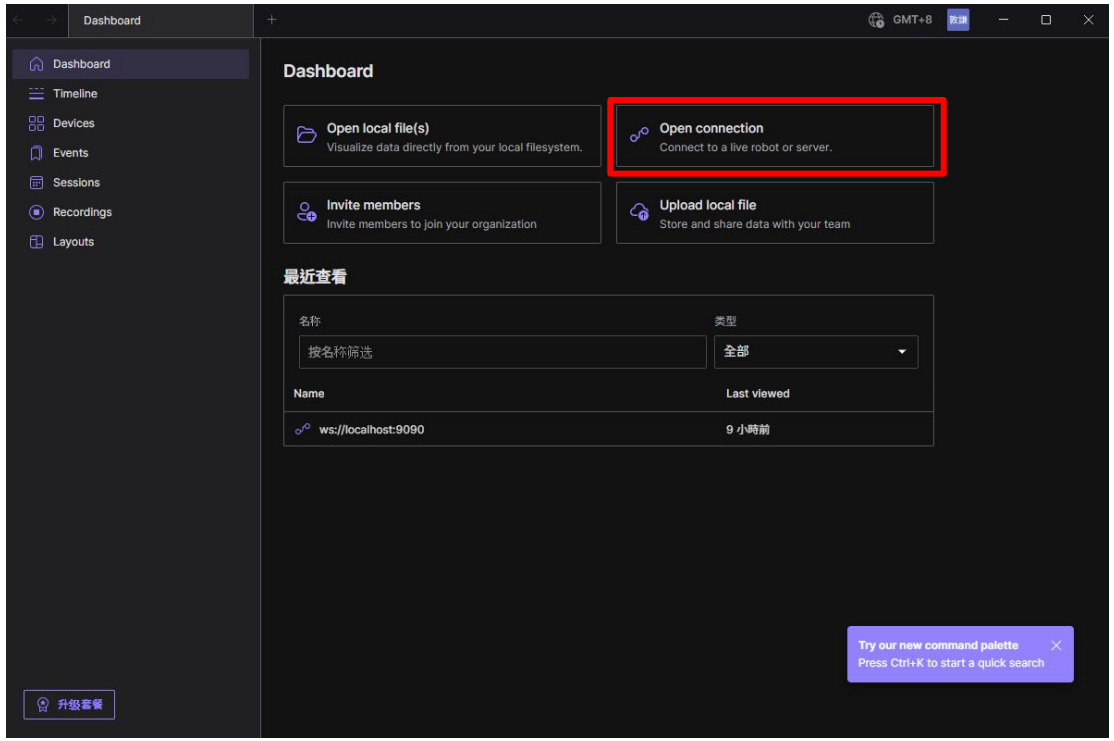
- Continue with “EASY” mode.
- Choose “FINAL PROJECT” scene. (You may need to wait for a while.)
- Switch from “MANUAL” (Unity) to “AI” (ROS) mode.

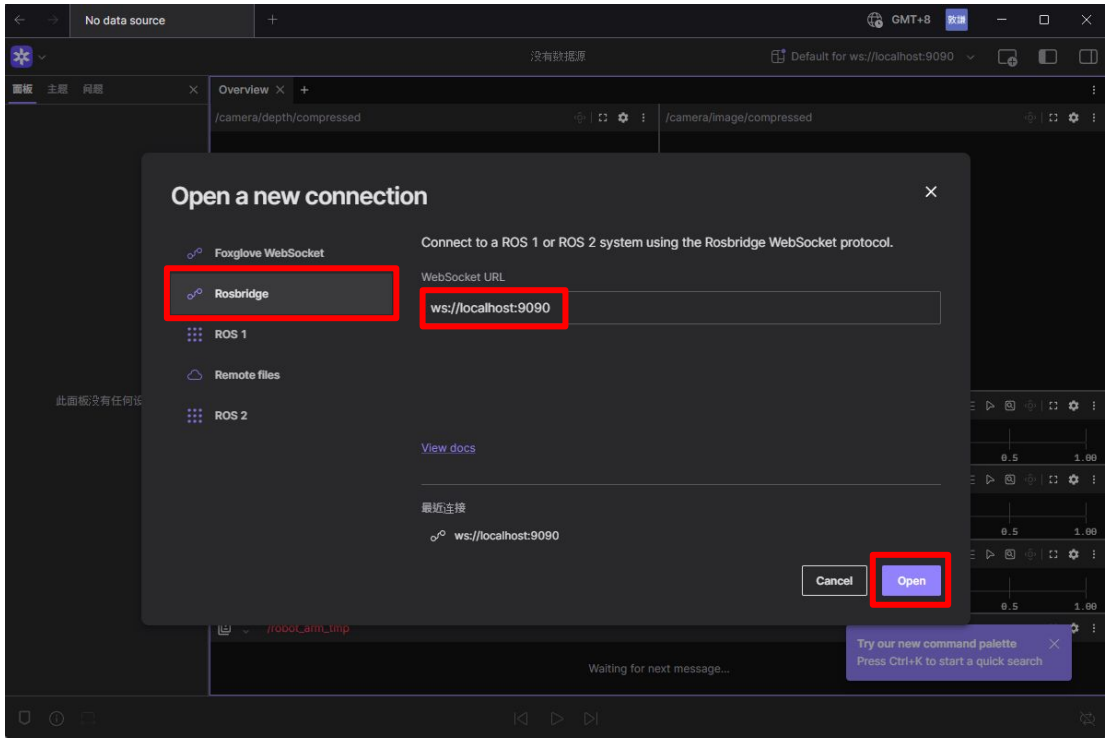
Data Collection

- Hold the **right mouse button** to change the view direction.
  Press the **WASD keys** to move the viewpoint.
  Use the **mouse wheel** to zoom in and out.
- Click “FINAL PROJECT” tab to reset the scene. (Scene will generated randomly.)
- You can turn off the LiDAR visualization if needed.

Data Collection

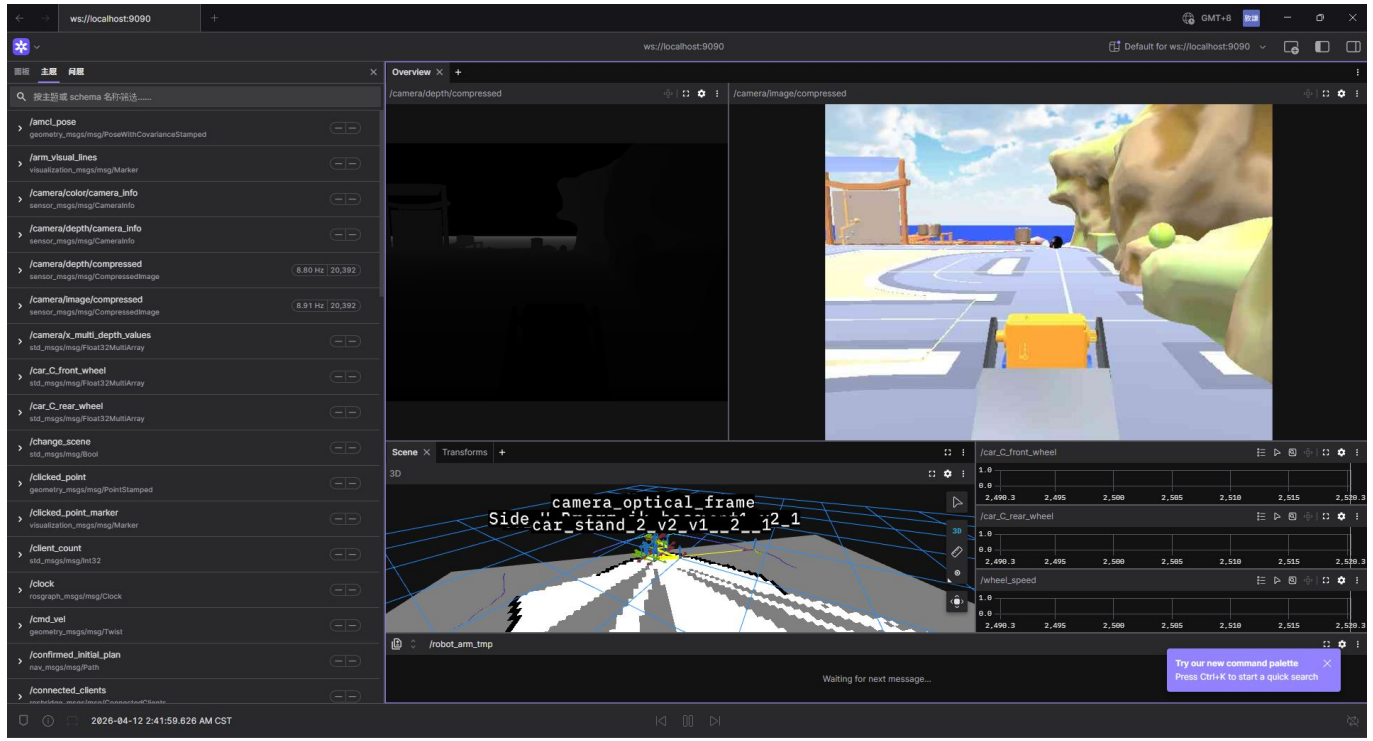
- Open Foxglove and connect to Rosbridge





Data Collection

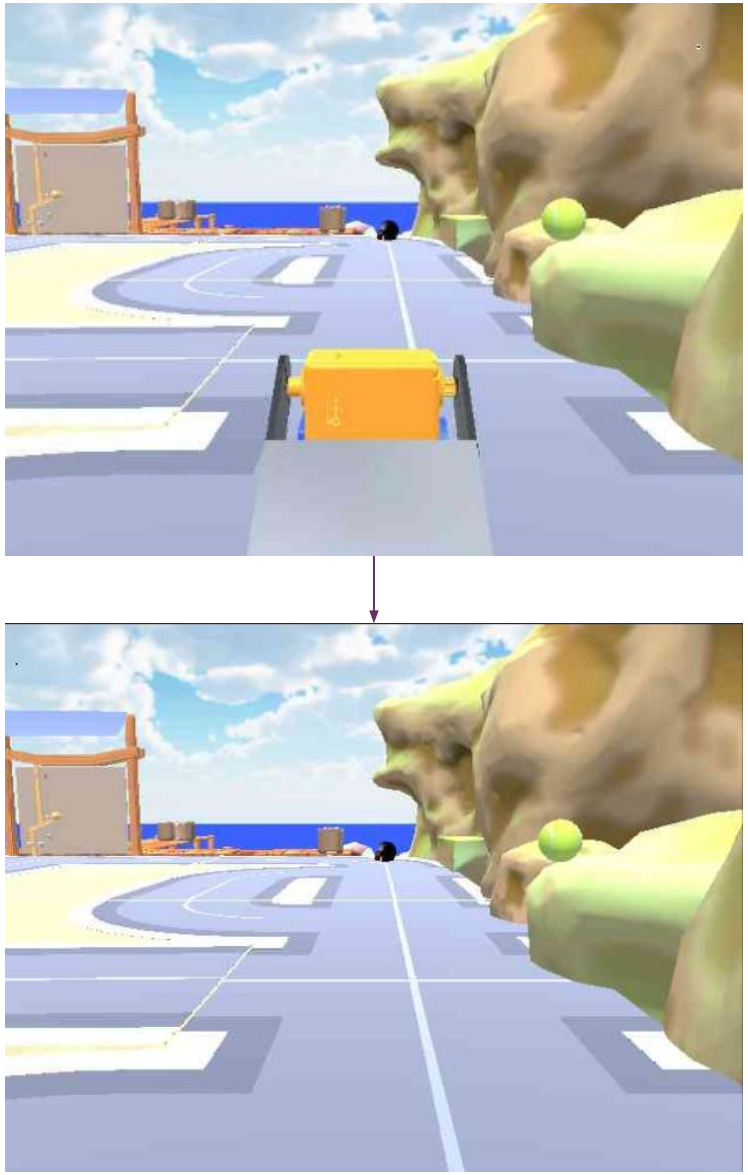
- You will see the image captured by the camera of the car.



Data Collection

- Execute pros_car

```
#Open another terminal
cd ./workspace/pros/pros_car/
./car_control.sh
r
ros2 run pros_car_py robot_control
#Select "Manual Arm Control"
#Select "[0]"
#Press b to reset (see the images on the right)
#Press q to quit
#Select "Control Vehicle"
```



w: 前進	s: 後退	c: 儲存當前畫面
e: 左轉	r: 右轉	
z: 停止	q:停止並結束控制	

Images will be stored at ./workspace/pros/pros_car/src/pros_car_py/images
Note that the GitHub repository already includes one image

Data Collection

- Adjust the moving and rotation speed here.

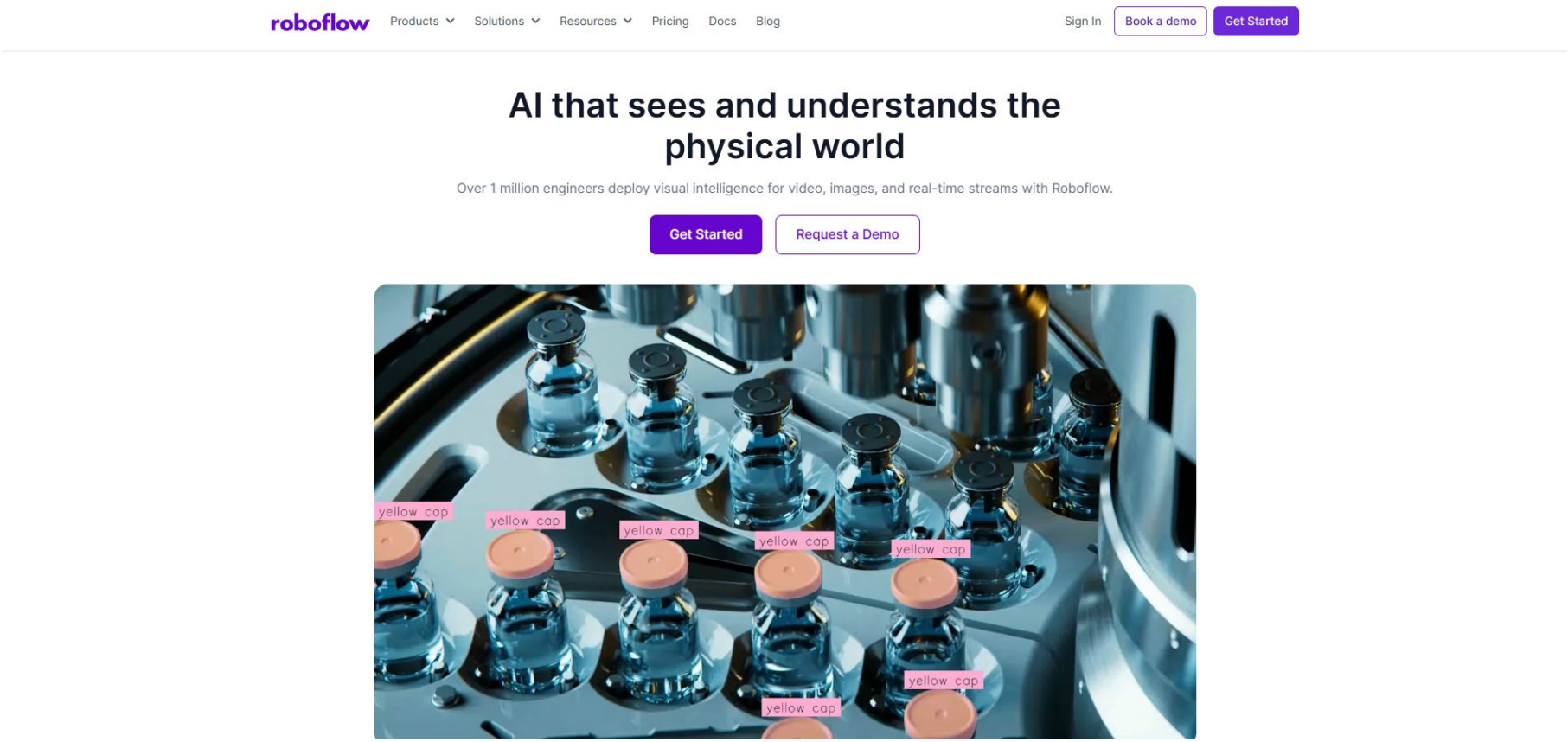
ros_communicator_config.py

workspace > pros > pros_car > src > pros_car_py > pros_car_py > ros_communicator_config.py

```python
# -*- coding: utf-8 -*-

# config.py

"""
vel, rotate_vel為自走車PID數值, 可於arduino程式碼查看

於ros_receive_and_data_processing/AI_node.py使用

前左、前右、後左、後右
"""
speed_ratio = 50
vel = 6.0*speed_ratio
vel_slow = 3.0*speed_ratio
rotate_vel = 6.0*speed_ratio
rotate_vel_slow = 5.0*speed_ratio
rotate_vel_median = 5.5*speed_ratio
ACTION_MAPPINGS = {
    "FORWARD": [vel, vel, vel, vel],  # 前進
    "FORWARD_SLOW": [vel_slow, vel_slow, vel_slow, vel_slow],  # 前進
    "LEFT_FRONT": [rotate_vel, rotate_vel * 1.2, rotate_vel, rotate_ve
    "COUNTERCLOCKWISE_ROTATION": [
```

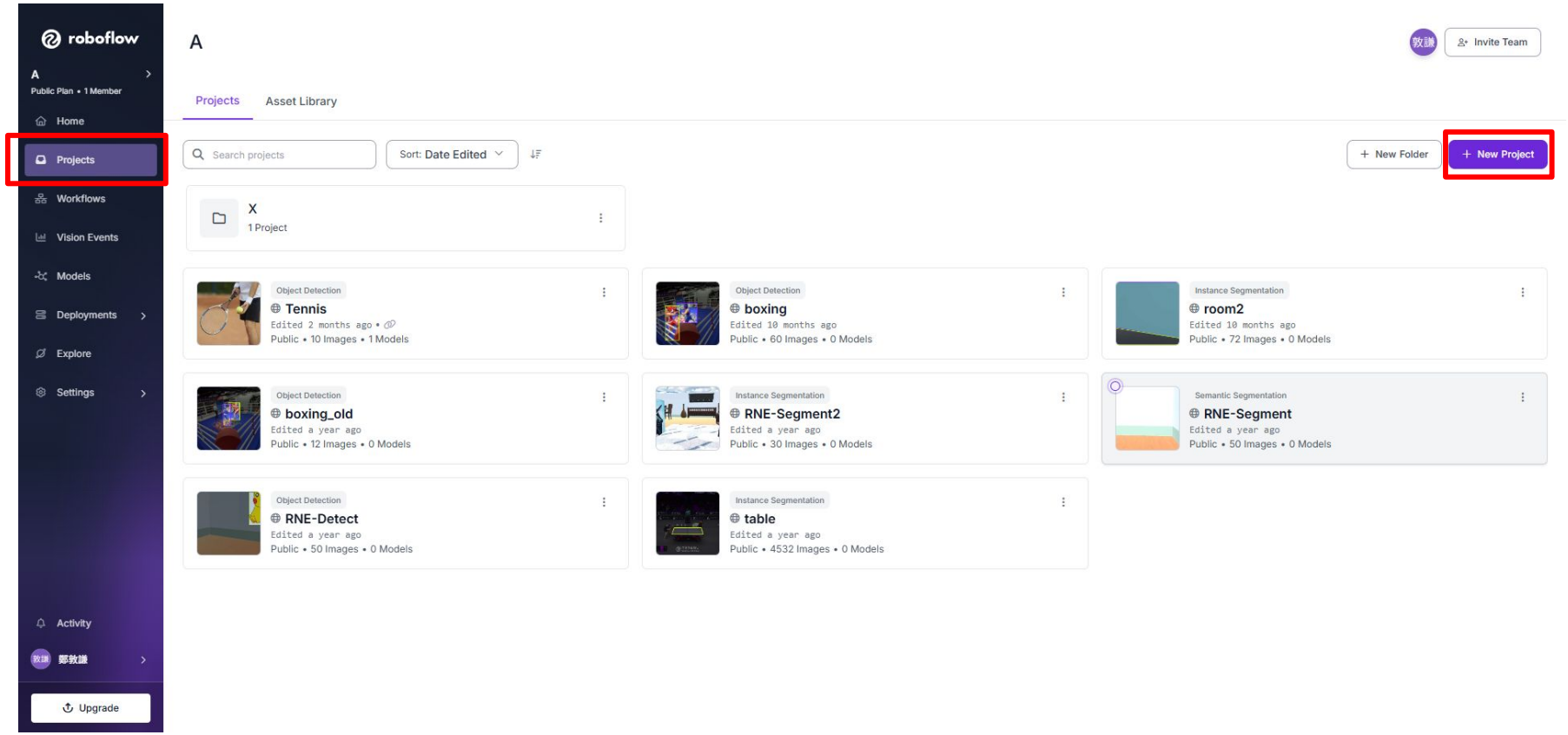
Data Labeling

Label Tool: Roboflow



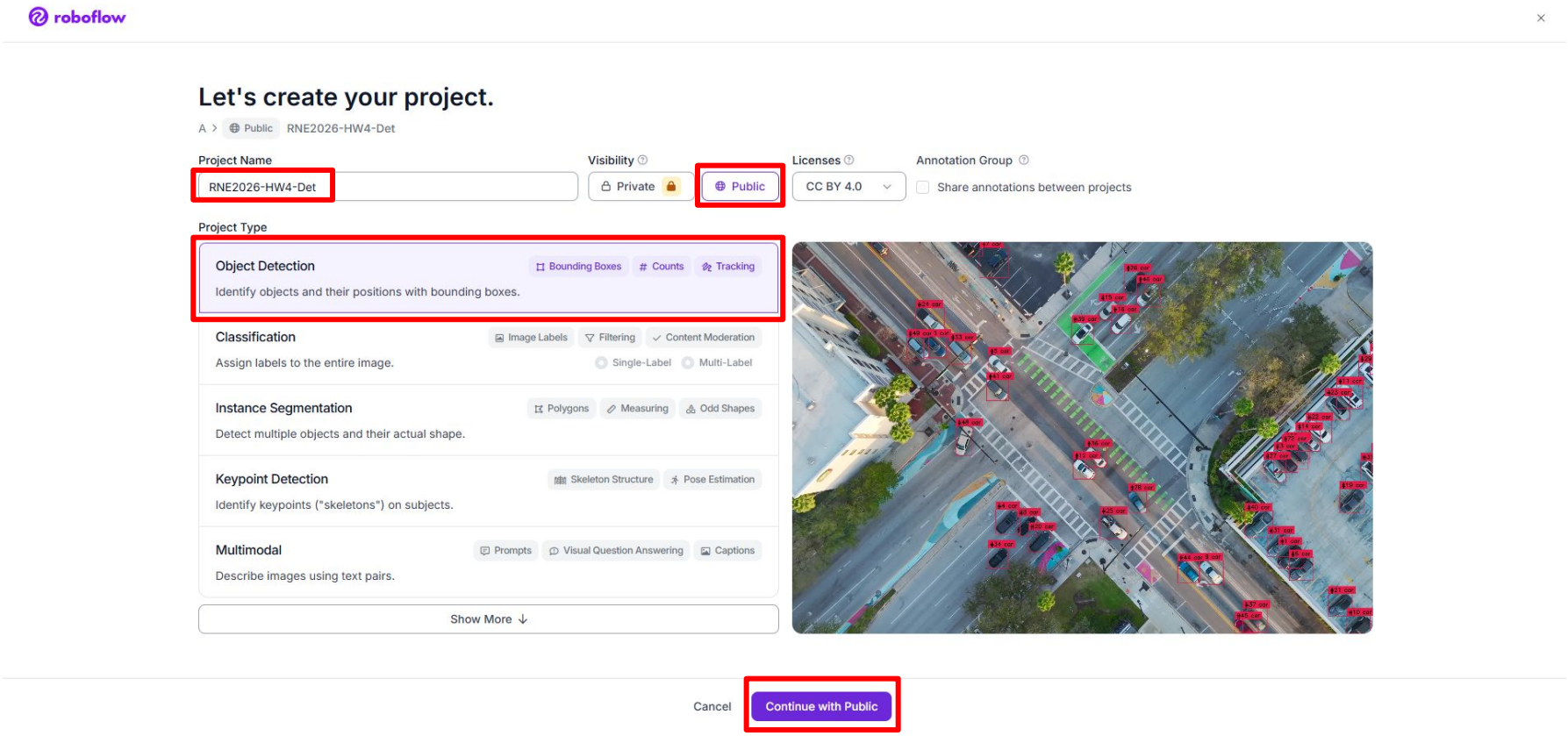
https://roboflow.com/

1. Create Project



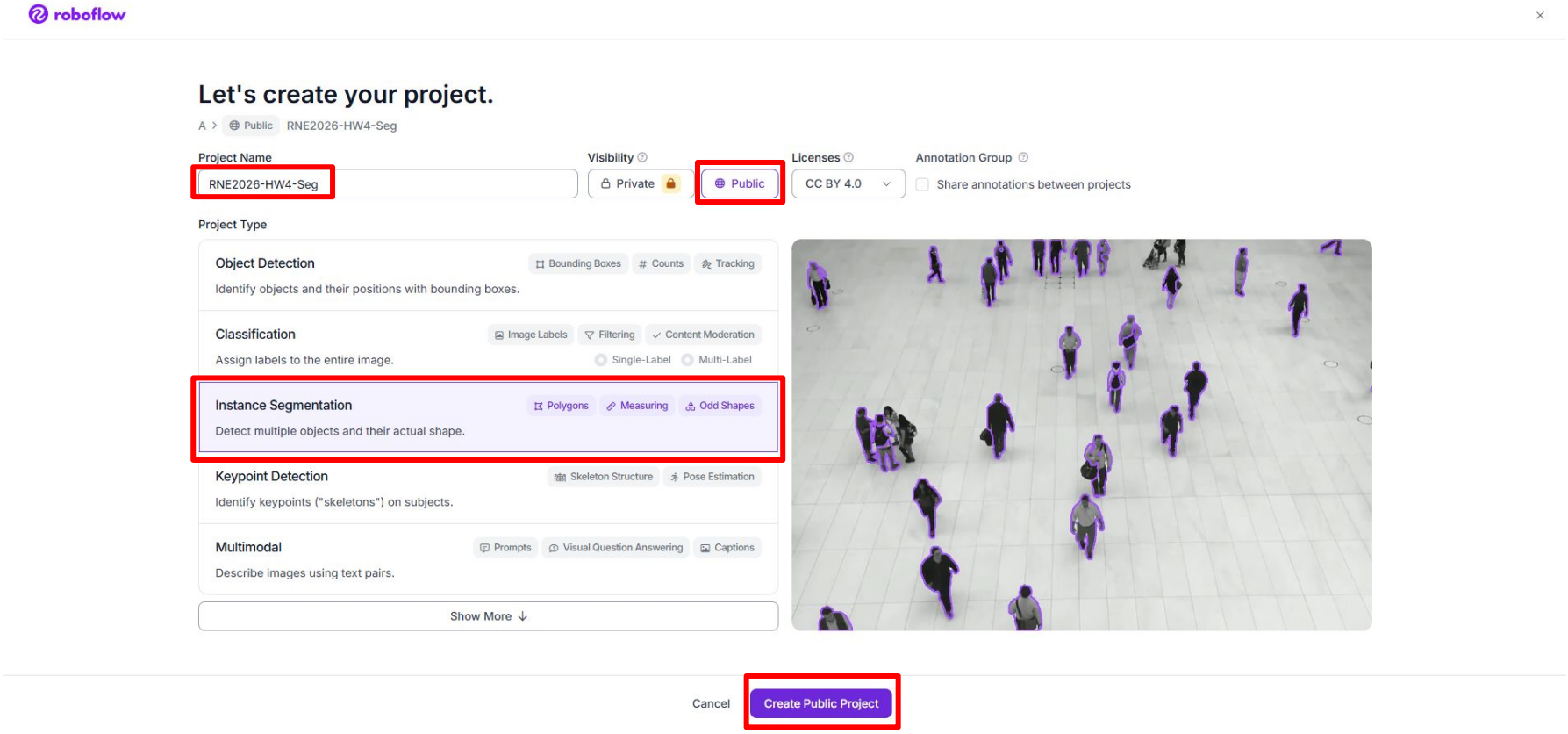
1. Create Project - Object Detection

- Use a random project name (e.g., “dfgyyuawegbfa“), so others cannot easily copy your homework.



1. Create Project - Semantic Segmentation

- Use a random project name (e.g., “dfgyyuawegbfa“), so others cannot easily copy your homework.



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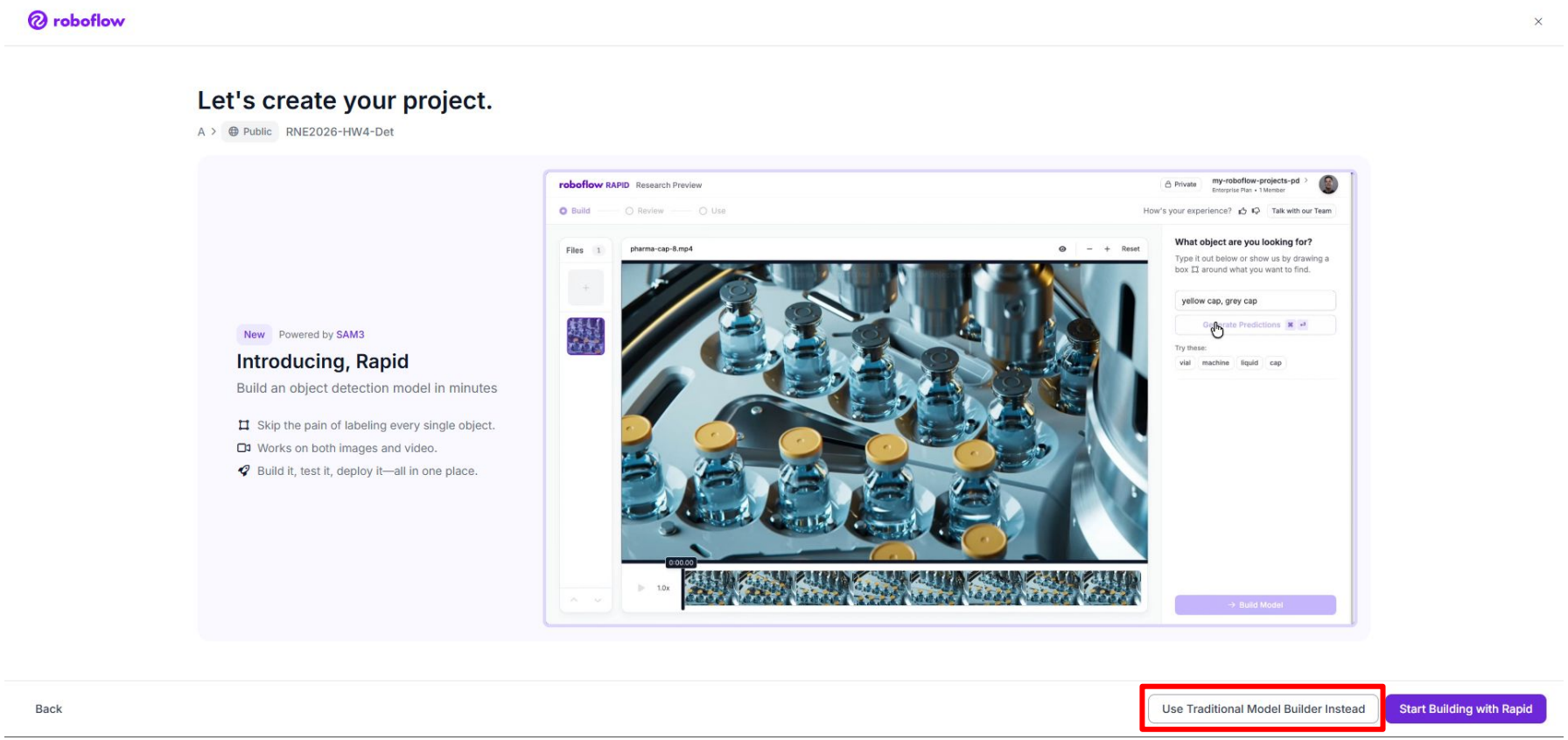
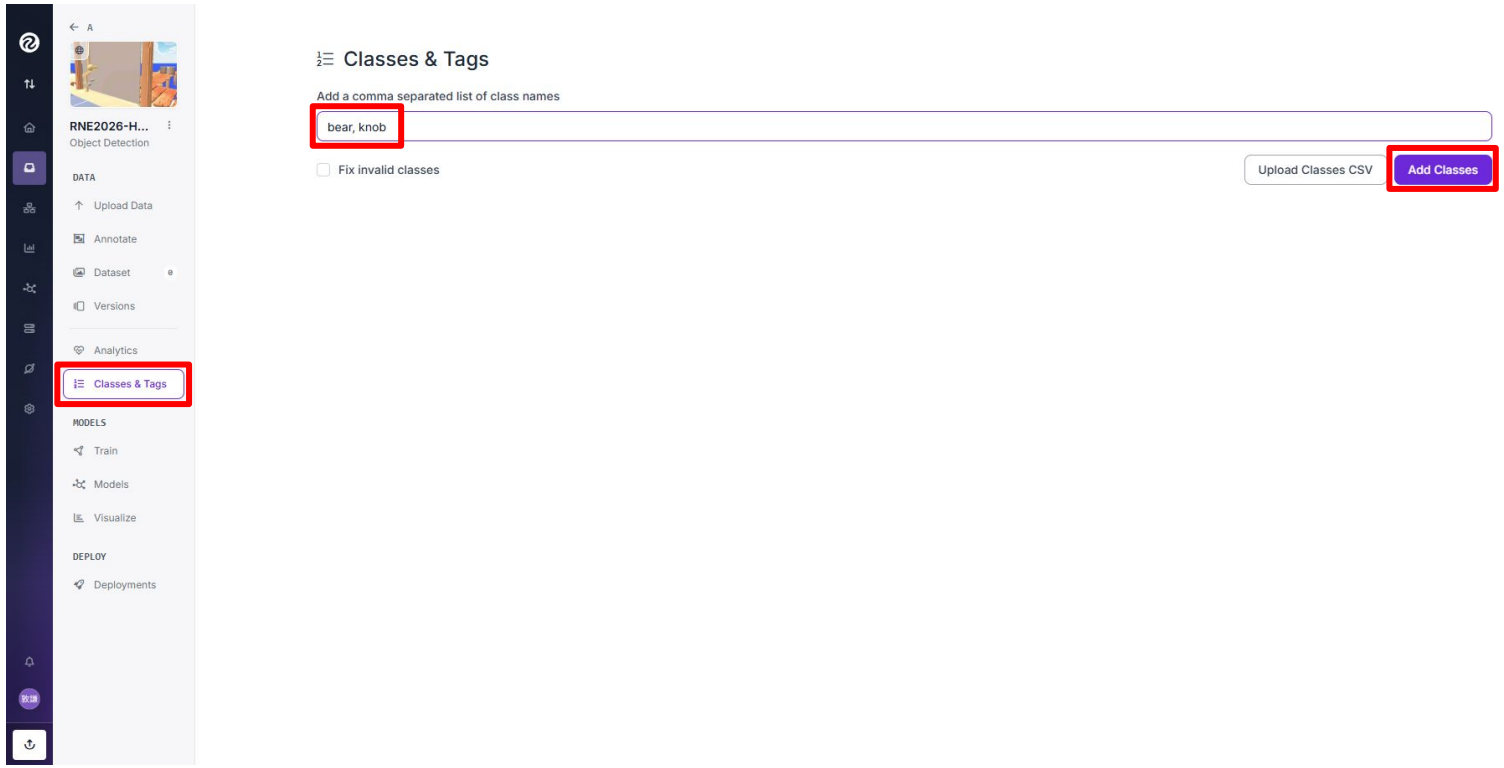
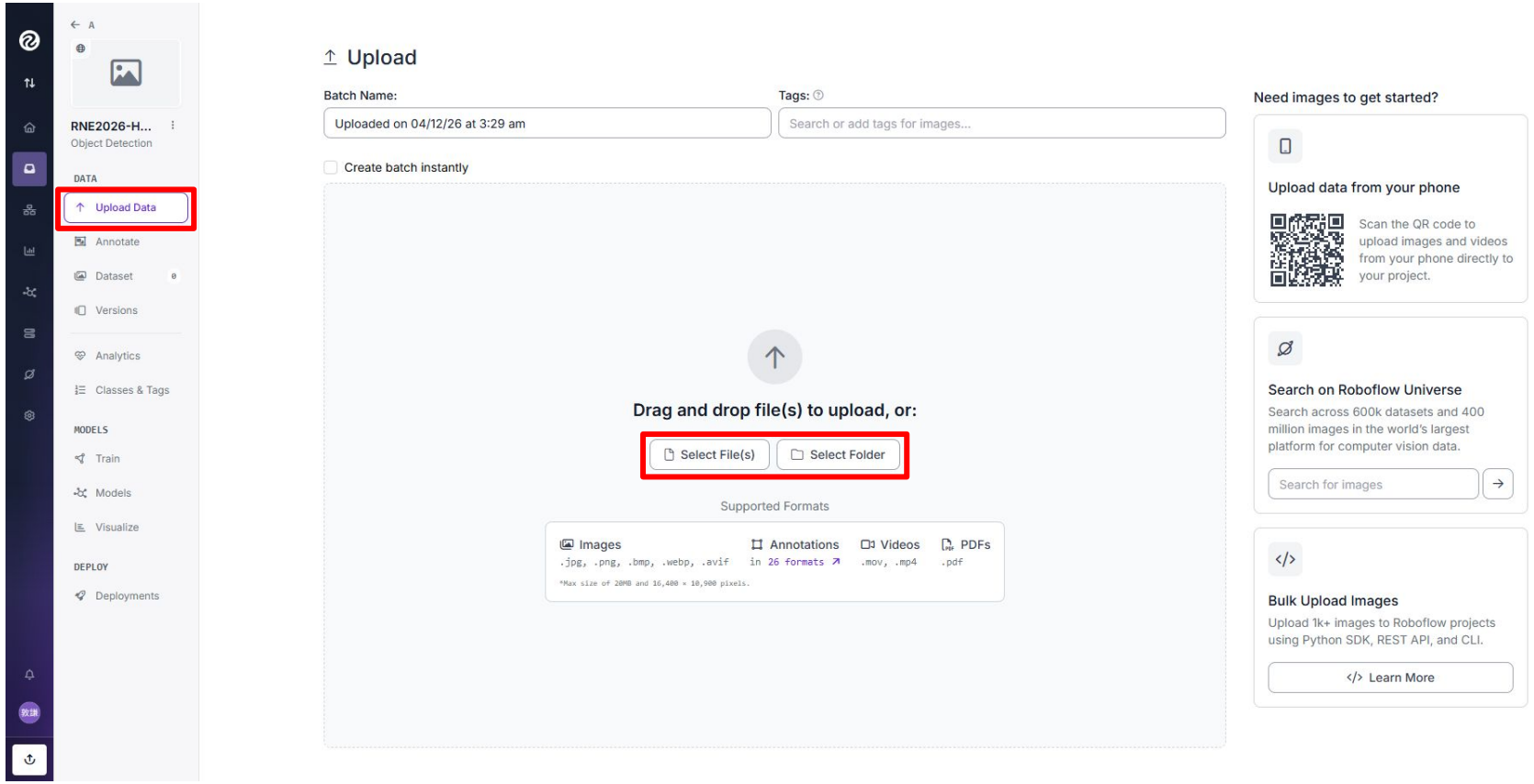


2. Define Classes

- Object Detection: bear, knob
- Semantic Segmentation: road, bridge



3. Upload Dataset



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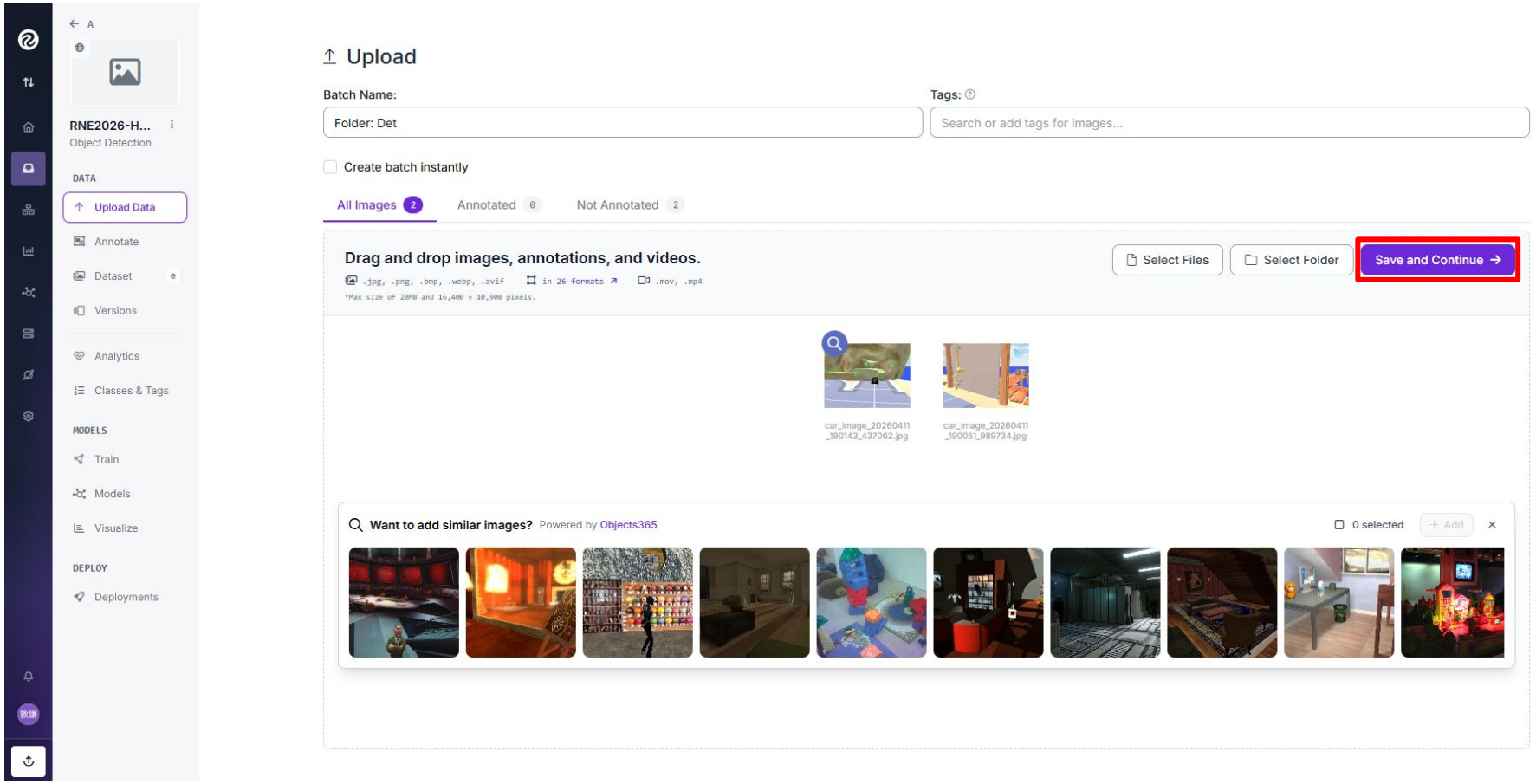
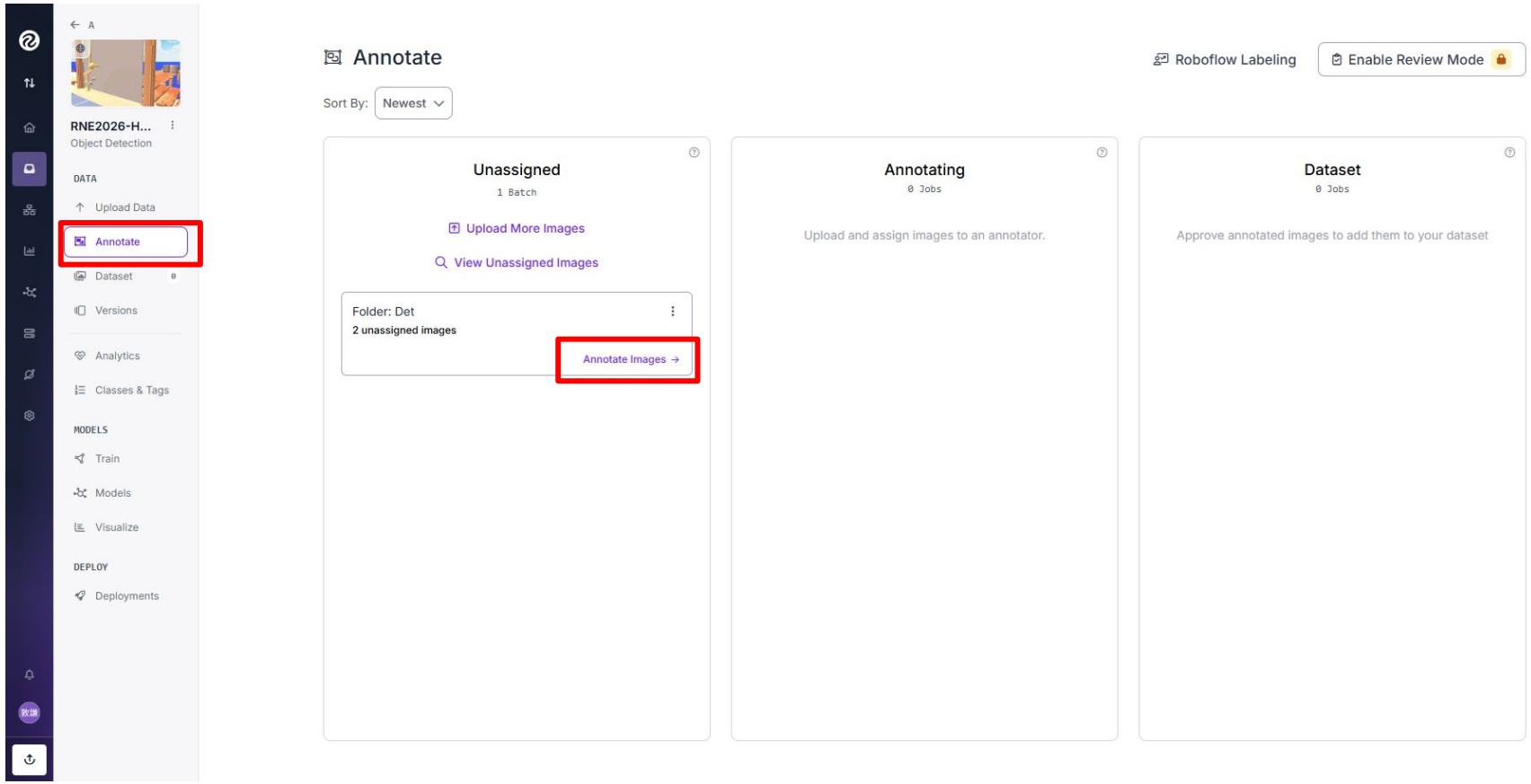


4. Annotate Images



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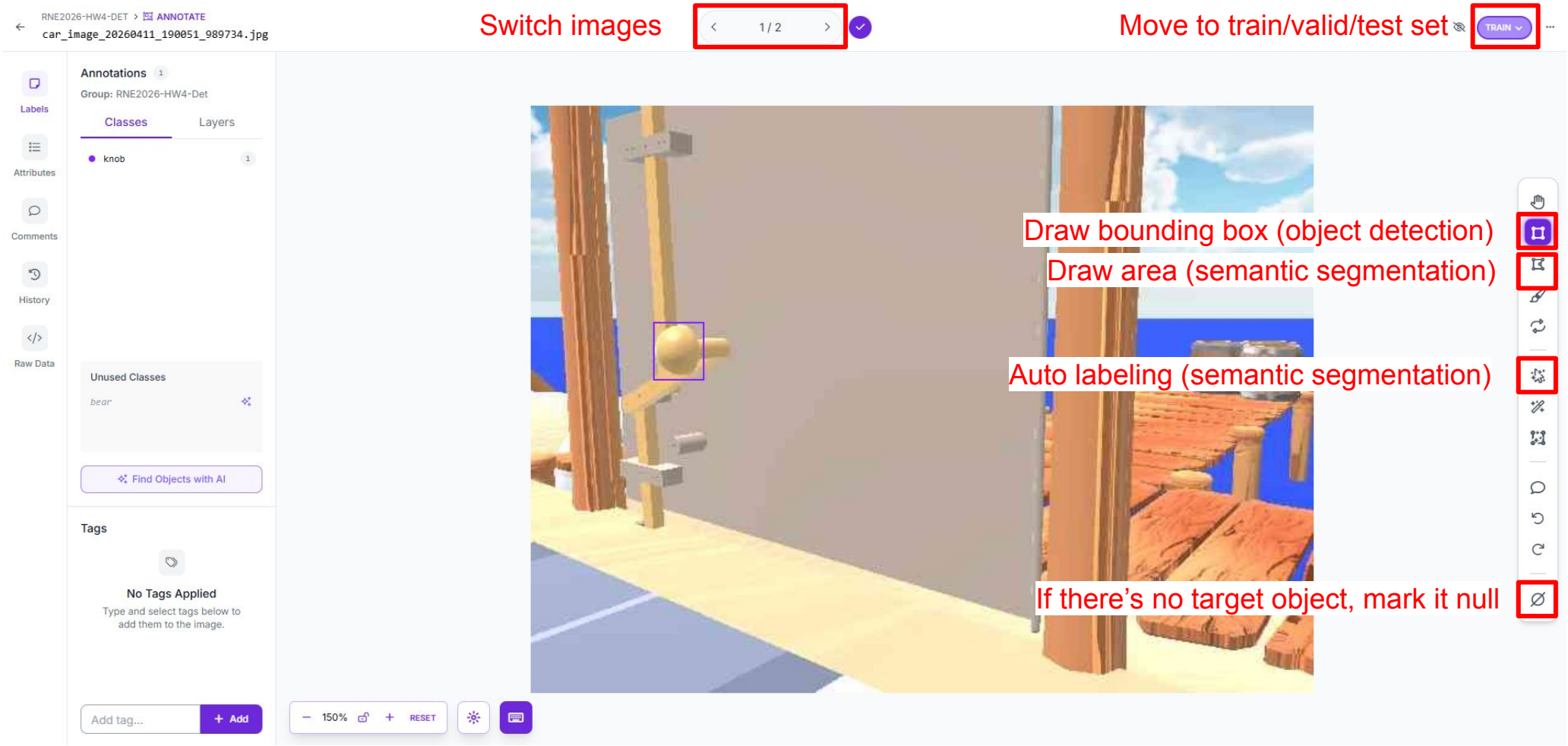

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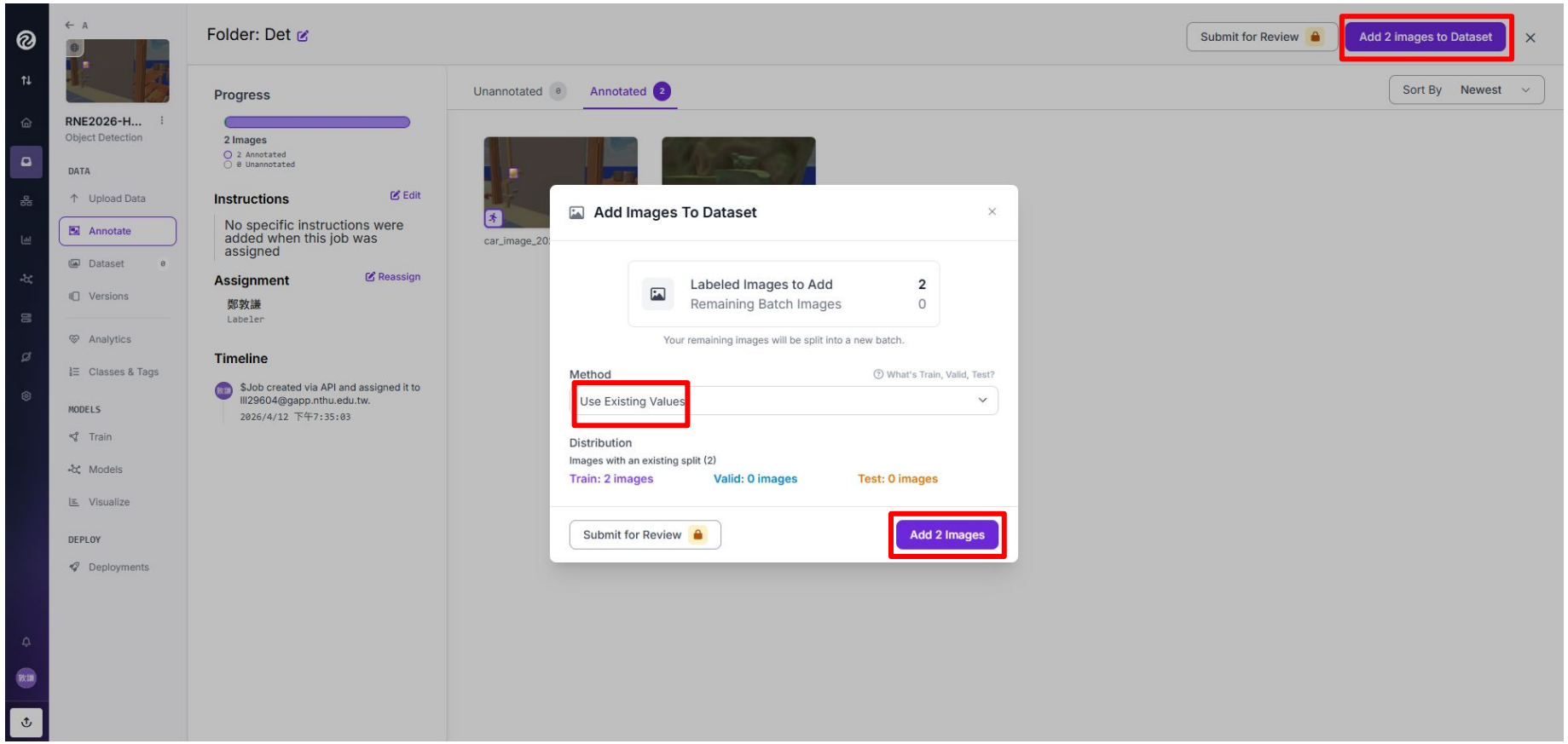


5. Export Dataset

- YOLO automatically performs image resizing and augmentation during training (See the [official documentation] for details)

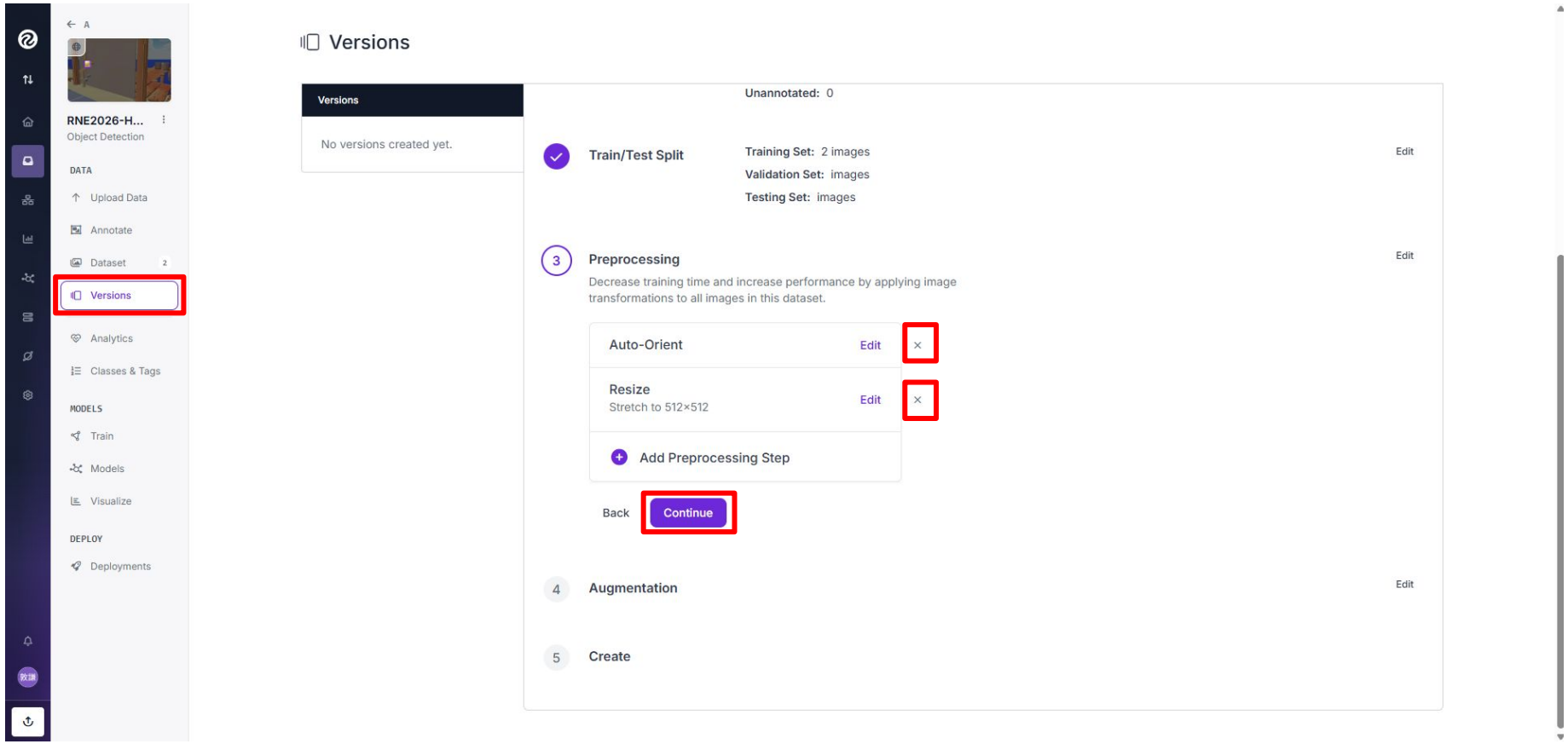
imgsz	int	640	Target image size for training. Images are resized to squares with sides equal to the specified value (if rect=False), preserving aspect ratio for YOLO models but not RT-DETR. Affects model accuracy and computational complexity.

Augmentation Settings and Hyperparameters

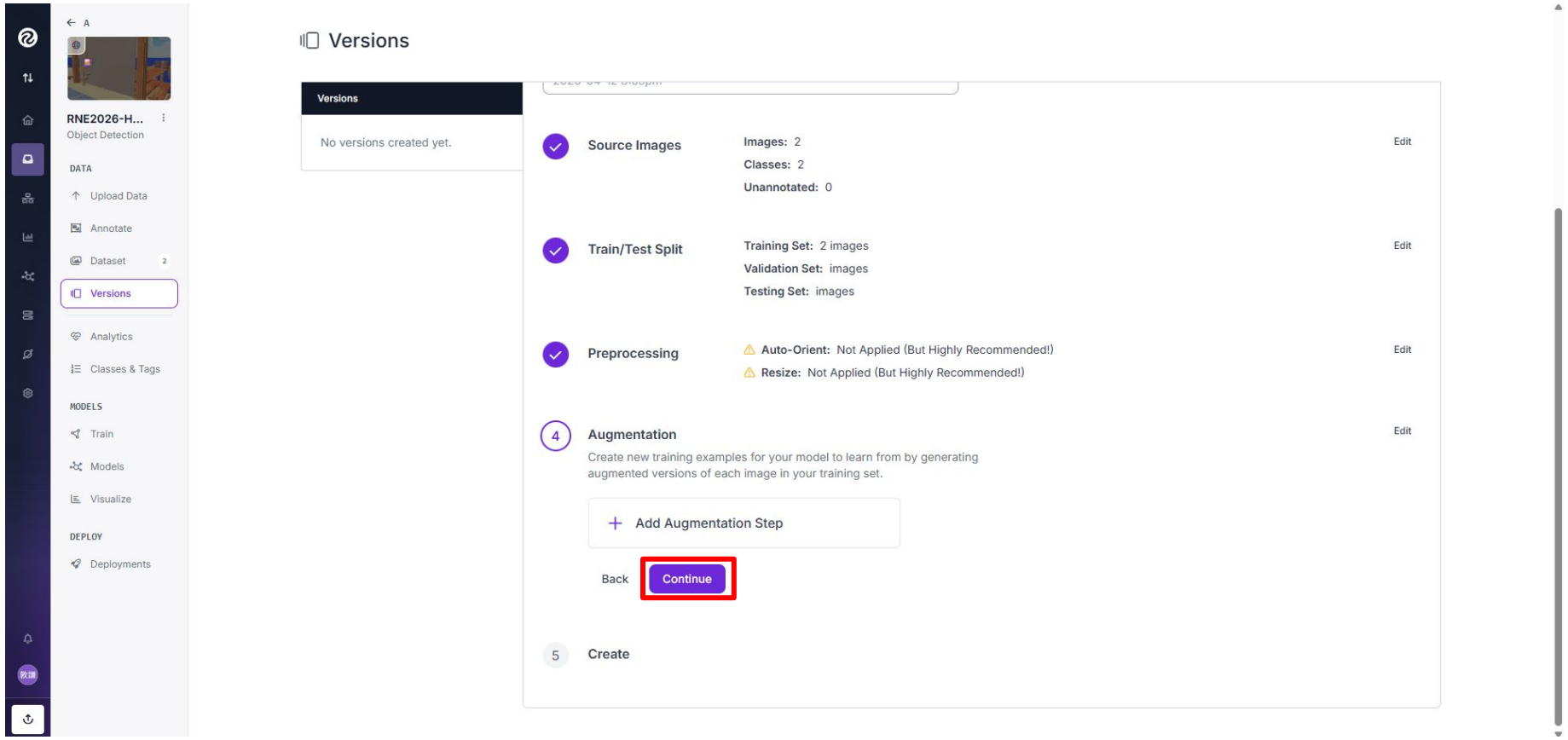
Augmentation techniques are essential for improving the robustness and performance of YOLO models by introducing variability into the training data, helping the model generalize better to unseen data. The following table outlines the purpose and effect of each augmentation argument:

Argument	Type	Default	Supported Tasks	Range	Description
hsv_h	float	0.015	detect, segment, pose, obb, classify	0.0 - 1.0	Adjusts the hue of the image by a fraction of the color wheel, introducing color variability. Helps the model generalize across different lighting conditions.
hsv_s	float	0.7	detect, segment, pose, obb, classify	0.0 - 1.0	Alters the saturation of the image by a fraction, affecting the intensity of colors. Useful for simulating different environmental conditions.

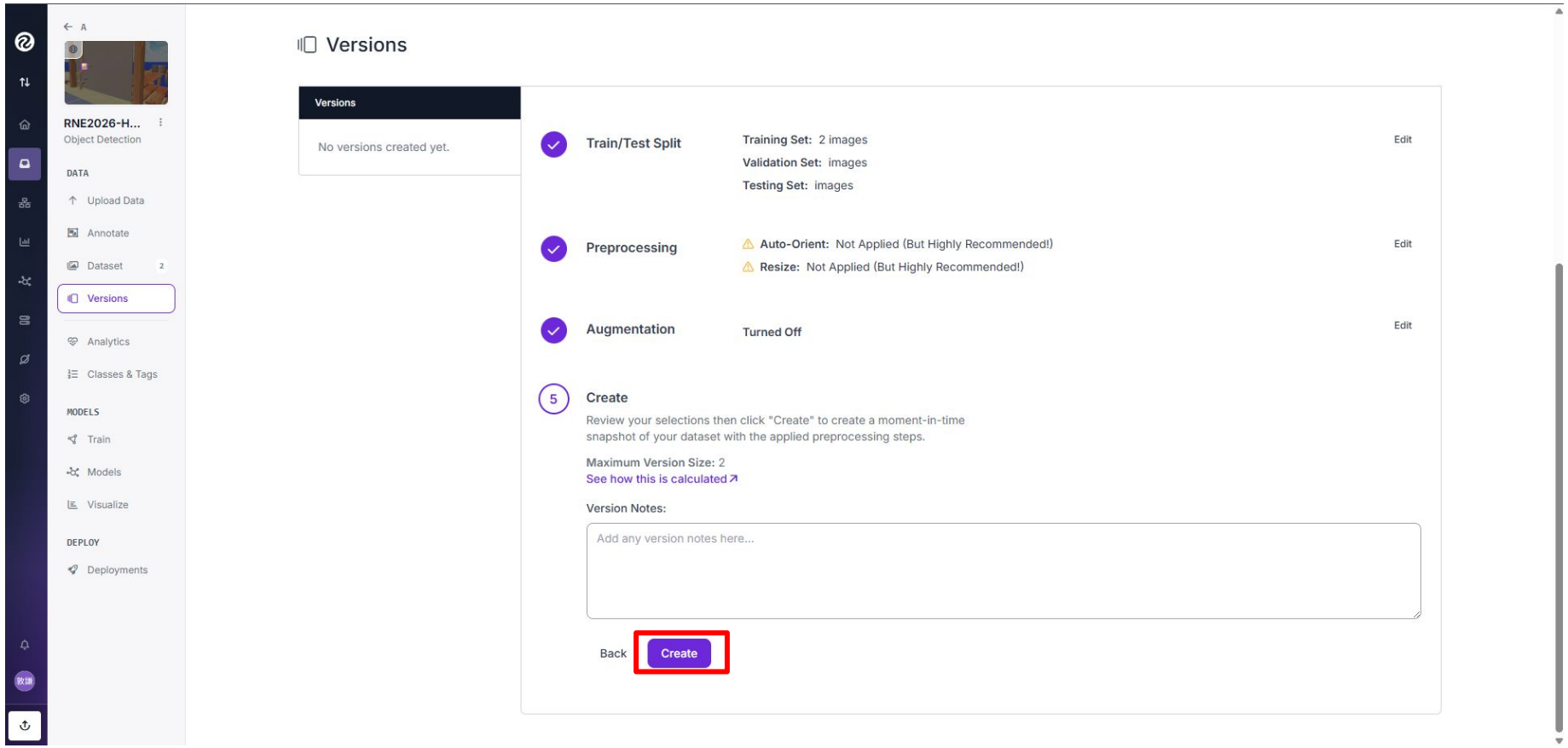
5. Export Dataset



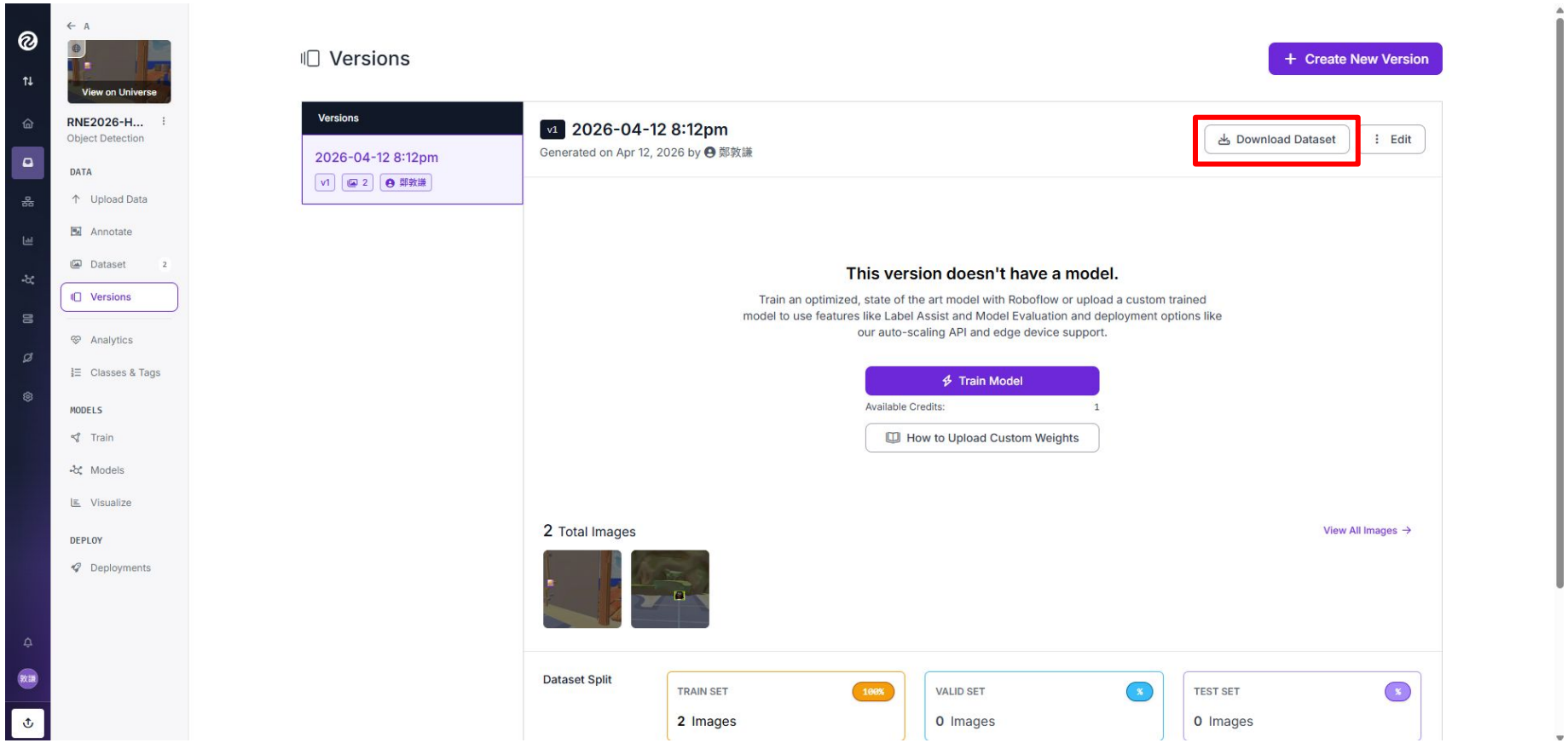
5. Export Dataset



5. Export Dataset



5. Export Dataset



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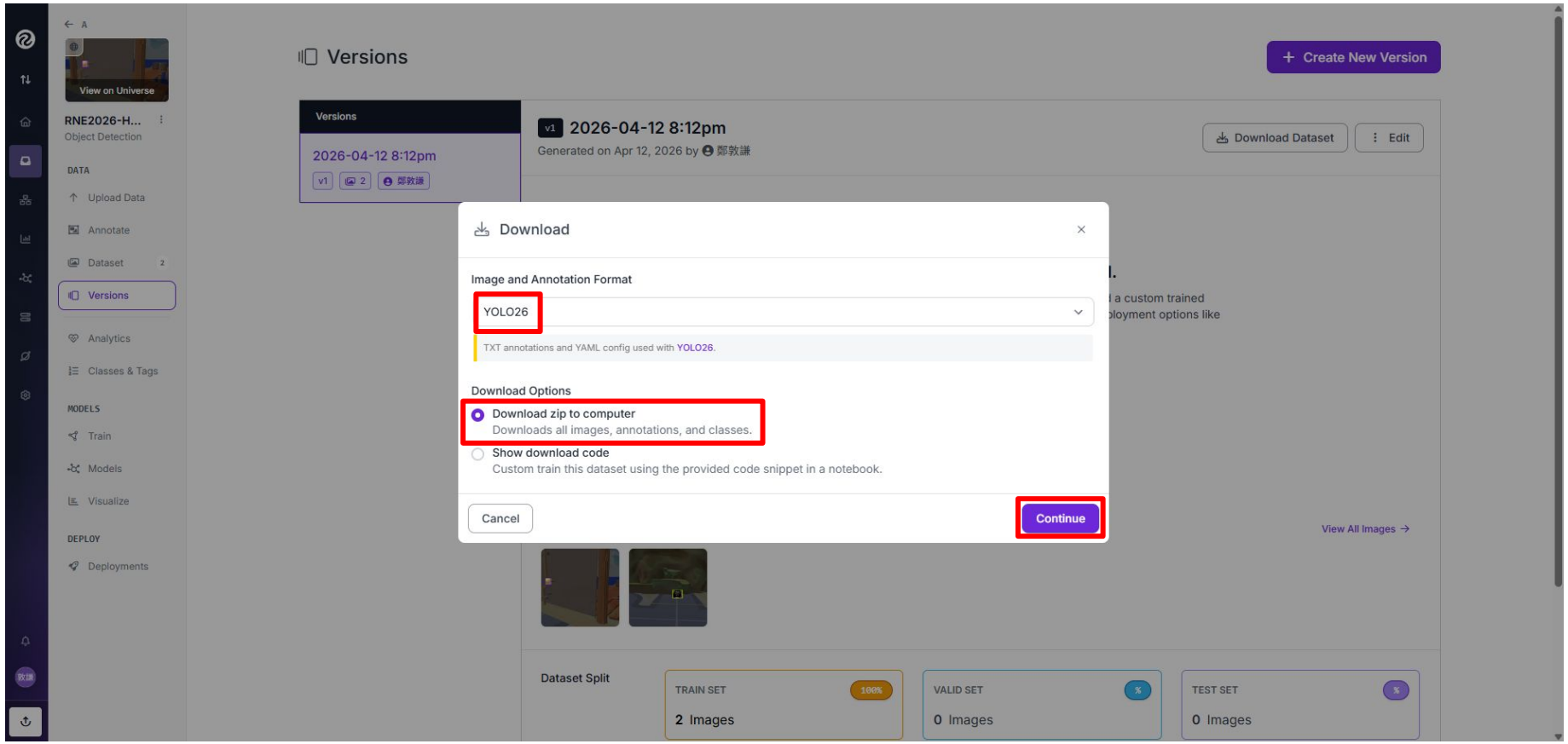


YOLO Training

Test CUDA Version

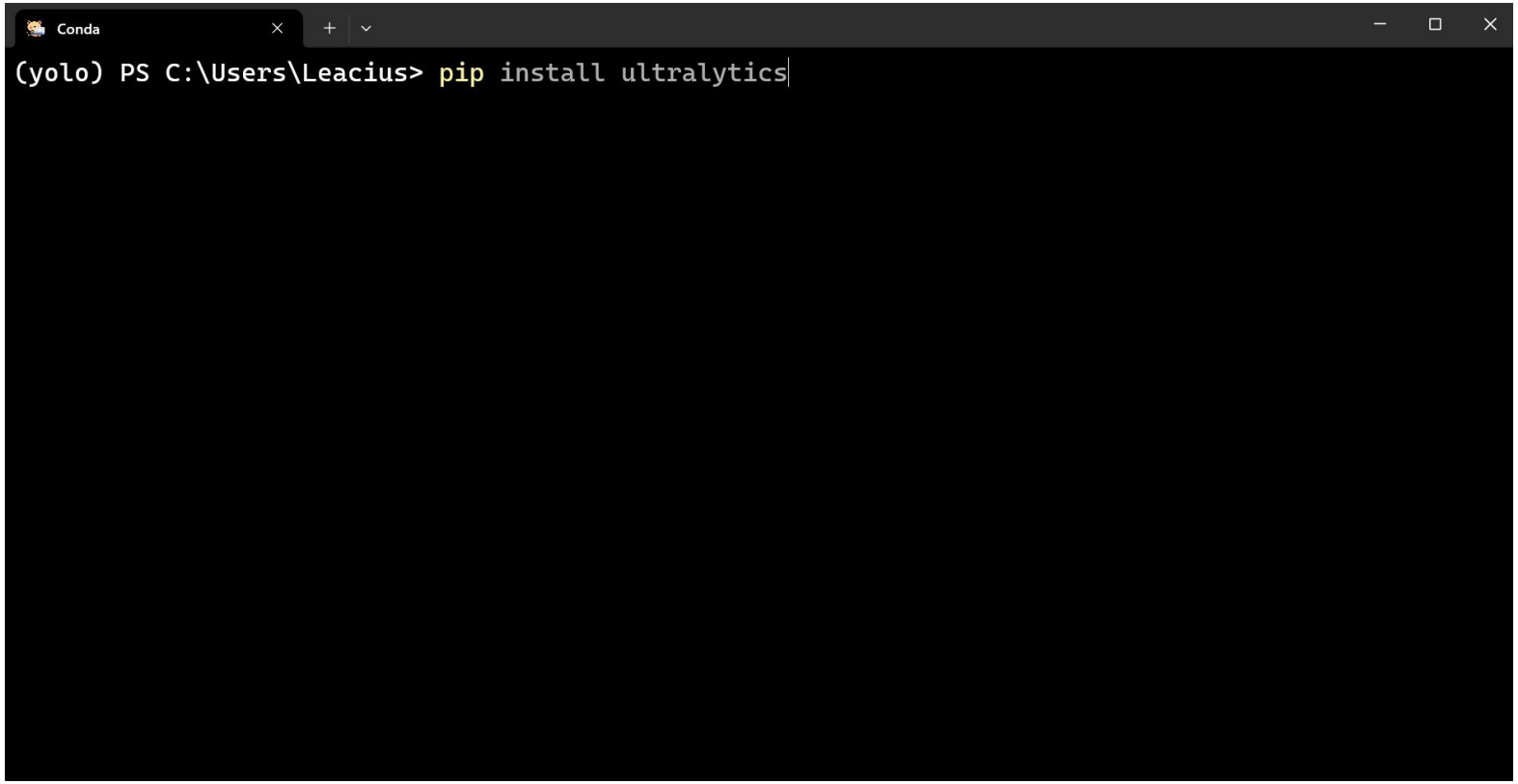
```
(YOLO) PS C:\Users\louis\Desktop> nvidia-smi
Sun Apr 12 22:38:35 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 591.86                 Driver Version: 591.86         CUDA Version: 13.1     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                  Driver-Model | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA GeForce RTX 5070      WDDM  |   00000000:02:00.0  On |                  N/A |
| 30%   47C    P0             61W /  250W |    2746MiB /  12227MiB |     61%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+
```

Install Pytorch

PyTorch Build	**Stable (2.11.0)**		Preview (Nightly)		
Your OS	Linux		Mac	**Windows**	
Package	**Pip**		**LibTorch**	Source	
Language	**Python**		C++ / Java		
Compute Platform	**CUDA 12.6**	CUDA 12.8	CUDA 13.0	~~ROCm 7.2~~	CPU
Run this Command:	`pip3 install torch torchvision --index-url https://download.pytorch.org/whl/cu126`				

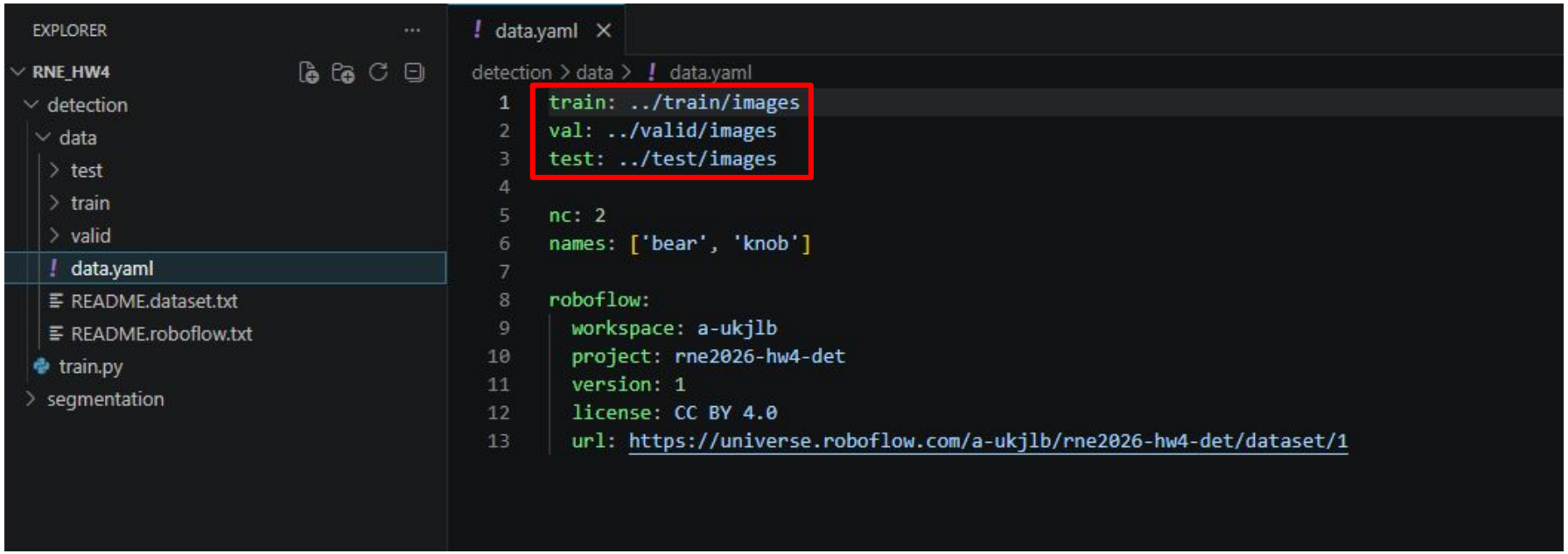
https://pytorch.org/

Install Ultralytics



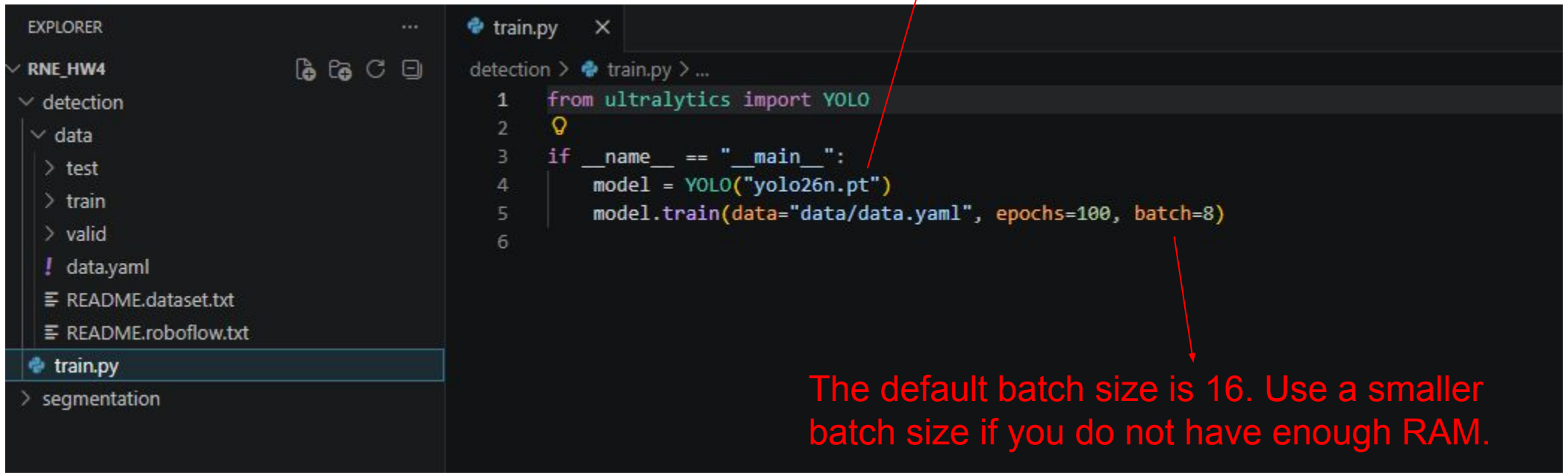
YOLO Configs

Modify to absolute path



YOLO Training

Specify the model name.
Object Detection: yolo26n.pt, yolo26s.pt, …
Semantic Segmentation: yolo26n-seg.pt, yolo26s-seg.pt, …



The default batch size is 16. Use a smaller batch size if you do not have enough RAM.

Execute “cd detection; python train.py”
The best model weight will be stored at detection\runs\detect\train\weights\best.pt

YOLO Testing

Test Your YOLO Model

- Put your model (.pt file) in ./workspace/pros/ros2_yolo_integration/src/yolo_example_pkg/models
- Modify the model path in workspace/pros/ros2_yolo_integration/src/yolo_example_pkg/yolo_example_pkg/object_detect.py (line 26)
- Execute ros2_yolo_integration

```
cd ./workspace/pros/ros2_yolo_integration/
./yolo_activate.sh
r
ros2 run yolo_example_pkg yolo_node
```

- See the result in Foxglove

Grading & Submission

Grading

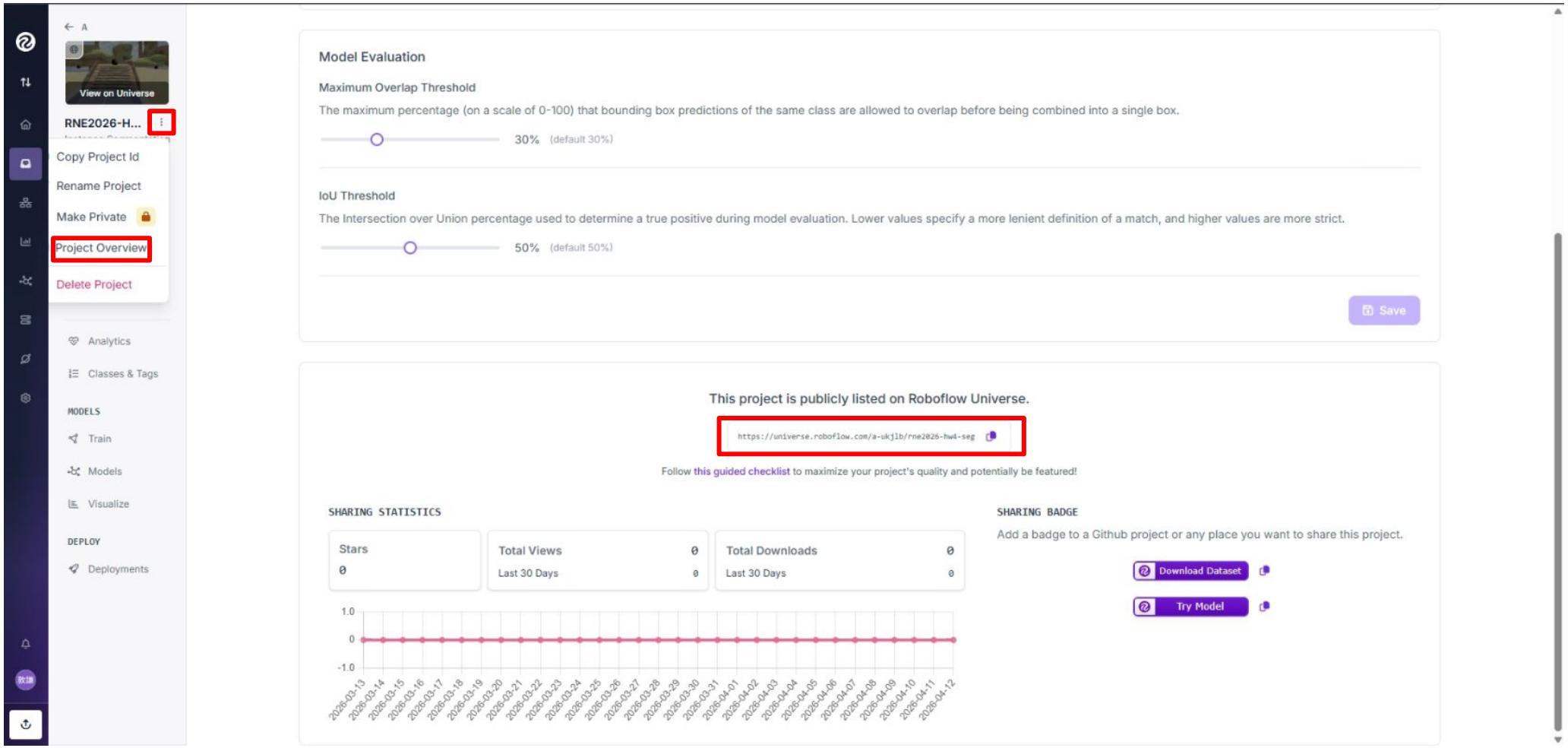
- Object Detection (45%)
  - Bear (20%) + Knob (20%)
  - Quality (5%)
- Semantic Segmentation (45%)
  - Road (20%) + Bridge (20%)
  - Quality (5%)
- Report (10%)

Report

- Two Roboflow project links
- Dataset configuration
  - # of images, **object categories**, train/test split, augmentation setting, …
- YOLO training configuration
  - model type, # of epochs, …
- **Description of dataset diversity**
  - text description, example images
- **Description of your navigation strategy in the final project**
  - text description, figures, algorithm

The report may be written in either English or Mandarin.

Roboflow Project Link



Submission

- Submit one zip file ({studentID}_HW4.zip), including:
  - Model * 2 (detection.pt & segmentation.pt)
  - Demo video * 2 (detection.mp4 & segmentation.mp4)
    - Record the detection results in Foxglove
    - Each video should be about 30~60 seconds long
    - These videos will be used for grading
  - Report (report.pdf)
- 10 points will be deducted for each missing file or incorrect filename
- **Deadline: 2026/05/03 23:59**
- No late submissions will be accepted, even in the case of project environment issues

HW4 Discussion